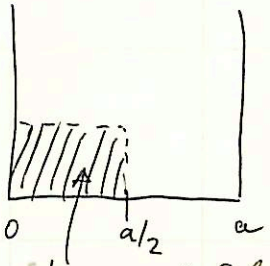
P2.8



(a) $t=0$ probability density

$|\Psi(x,0)|^2 dx = C\,dx$ = probability to find particle in the interval $(x, x+dx)$ at $t=0$ (for $0 \le x \le a/2$).

Total probability $= 1 \Rightarrow \int_0^{a/2} dx\, C = 1 = \frac{a}{2} C$, $C = \frac{2}{a}$

$\Rightarrow$ choosing $\Psi(x,0)$ = real we find

$$\Psi(x,0) = \sqrt{\frac{2}{a}}$$

(b) from $\Psi(x,0) = \sum_{n=1}^{\infty} c_n \psi_n(x)$ follows $\Psi(x,t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{-iE_n t/\hbar}$ at later times.

Probability of an energy measurement to yield $E_1 = \frac{\pi^2 \hbar^2}{2ma^2}$ is $|c_1|^2$.

But $c_1 = \int_0^a dx\, \psi_1^*(x)\, \Psi(x,0) = \int_0^{a/2} dx \sqrt{\frac{2}{a}} \cdot \sqrt{\frac{2}{a}} \sin\frac{\pi x}{a}$

$$= \frac{2}{a} \int_0^{\pi/2} d\left(\frac{x\pi}{a}\right) \frac{a}{\pi} \sin\left(\frac{\pi x}{a}\right) = \frac{2}{\pi} \int_0^{\pi/2} dy \sin y = \frac{2}{\pi} (-\cos y)\Big|_0^{\pi/2} = \frac{2}{\pi}.$$

$$\Rightarrow P_1 = \left(\frac{2}{\pi}\right)^2.$$

P2.9 $\Psi(x,0) = A\, x(a-x) \quad (0 \le x \le a)$

The normalization constant $A = \sqrt{\frac{30}{a^5}}$ is found in Example 2.2.

Calculate

$$\langle H \rangle = \int \Psi^*(x,0)\, \hat{H}\, \Psi(x,0)\, dx =$$

$$= A^2 \int_0^a x(a-x) \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \right) x(a-x)\, dx =$$

$$= -\frac{\hbar^2}{2m} A^2 \int_0^a x(a-x) \underbrace{\frac{d^2}{dx^2}(ax - x^2)}_{\frac{d}{dx}(a-2x) = -2} dx = \frac{\hbar^2}{m} A^2 \int_0^a (xa - x^2)\, dx$$

$$= \frac{\hbar^2}{m} A^2 \left(a\frac{x^2}{2} - \frac{x^3}{3} \right) \Big|_0^a = \frac{\hbar^2}{m} \frac{30}{a^5}\, a^3 \underbrace{\left(\frac{1}{2} - \frac{1}{3}\right)}_{1/6} = \frac{5\hbar^2}{m a^2}$$

In agreement with Example 2.3.

P2.10 (a) $\psi_n = \frac{1}{\sqrt{n!}}(a_+)^n \psi_0$

$$\Rightarrow \psi_2 = \frac{1}{\sqrt{2}}(a_+)^2\psi_0 = \frac{1}{\sqrt{2}}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\frac{1}{2\hbar m\omega}\left(-ip+m\omega x\right)^2 e^{-\frac{m\omega}{2\hbar}x^2}$$

$$= \frac{1}{2\sqrt{2}\hbar m\omega}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\left(-\hbar\frac{d}{dx}+m\omega x\right)\underbrace{\left(-\hbar\frac{d}{dx}+m\omega x\right)e^{-\frac{m\omega}{2\hbar}x^2}}_{\left(-\hbar\left(-\frac{m\omega}{2\hbar}\cdot 2x\right)+m\omega x\right)e^{-\frac{m\omega}{2\hbar}x^2}}$$

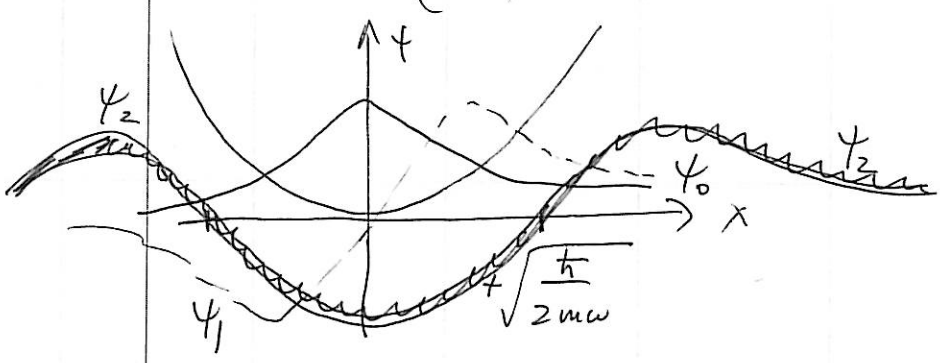
$$= \frac{1}{2\sqrt{2}\hbar m\omega}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\left(-\hbar\frac{d}{dx}+m\omega x\right)2m\omega\, x\, e^{-\frac{m\omega}{2\hbar}x^2}$$

$$= \frac{1}{\sqrt{2}\hbar}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\left[-\hbar-\hbar x\left(-\frac{m\omega}{2\hbar}\cdot 2x\right)+m\omega x\right]e^{-\frac{m\omega}{2\hbar}x^2}$$

$$= \frac{1}{\sqrt{2}\hbar}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\left(2m\omega x^2-\hbar\right)e^{-\frac{m\omega}{2\hbar}x^2} = \left(\frac{1}{\sqrt{2}} = \frac{1}{4^{1/4}}\right)$$

$$= \left(\frac{m\omega}{4\pi\hbar}\right)^{1/4}\left(2\frac{m\omega}{\hbar}x^2-1\right)e^{-\frac{m\omega}{2\hbar}x^2} = \psi_2(x).$$

(b) $\psi_1(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\sqrt{\frac{2m\omega}{\hbar}}\, x\, e^{-\frac{m\omega}{2\hbar}x^2}$; $\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}$



(c) $I_{01} = \int \psi_1^* \psi_0\, dx = 0$ because ψ_0 = even, ψ_1 = odd functions of x.

$I_{12} = 0$ (as well) $= \int \psi_2^* \psi_1 dx = 0$

$\Rightarrow$ Check $I_{02} = \int \psi_0^* \psi_2\, dx = \frac{1}{\sqrt{2}}\left(\frac{m\omega}{\pi\hbar}\right)^{1/2}\int_{-\infty}^{\infty} dx\left(2\frac{m\omega}{\hbar}x^2-1\right)e^{-\frac{m\omega}{\hbar}x^2}$

~~Step 4~~ Introduce

$$\xi = \sqrt{\frac{m\omega}{\hbar}}\, x \quad \Rightarrow \quad dx = \sqrt{\frac{\hbar}{m\omega}}\, d\xi$$

$$\Rightarrow I_{02} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\xi \, (2\xi^2 - 1)\, e^{-\xi^2}$$

We know that $J(\beta) = \int_{-\infty}^{\infty} d\xi \, e^{-\beta\xi^2} = \sqrt{\frac{\pi}{\beta}}$

because $J^2 = \int d\xi_1 \, d\xi_2 \, e^{-\beta(\xi_1^2 + \xi_2^2)} = 2\pi \int_0^{\infty} dr \cdot r \, e^{-\beta r^2} =$

$$= \pi \int_0^{\infty} d(r^2)\, e^{-\beta r^2} = \frac{\pi}{\beta}$$

Hence $\tilde{J}(\beta) = \int_{-\infty}^{\infty} d\xi \, \xi^2 e^{-\beta\xi^2} = -\frac{d}{d\beta} \int_{-\infty}^{\infty} d\xi \, e^{-\beta\xi^2} =$

$$= -\frac{d}{d\beta} \sqrt{\frac{\pi}{\beta}} = -\sqrt{\pi}\left(-\frac{1}{2}\right)\beta^{-3/2} = \frac{\sqrt{\pi}}{2\beta^{3/2}}$$

We need $I_{02} = \frac{1}{\sqrt{2\pi}} \left\{ 2 \int_{-\infty}^{\infty} d\xi \, \xi^2 e^{-\xi^2} - \int_{-\infty}^{\infty} d\xi \, e^{-\xi^2} \right\} =$

$$= \frac{1}{\sqrt{2\pi}} \left\{ 2\, \tilde{J}(\beta=1) - J(\beta=1) \right\} = \frac{1}{\sqrt{2\pi}} \left(2 \cdot \frac{\sqrt{\pi}}{2} - \sqrt{\pi} \right) = 0.$$

$$\Rightarrow \int \psi_2^* \psi_0 \, dx = \int \psi_0^* \psi_2 \, dx = 0 \quad \text{(as it should)}.$$

P2.11

$$\psi_0(x) = \alpha\, e^{-\xi^2/2}, \quad \psi_1(x) = \sqrt{2}\,\alpha\,\xi\, e^{-\xi^2/2}$$

(a) $\xi = \sqrt{\frac{m\omega}{\hbar}}\, x, \quad \alpha = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}$

$$\langle x\rangle = \sqrt{\frac{\hbar}{m\omega}}\,\langle \xi\rangle = 0 \quad \text{because} \quad \int_{-\infty}^{\infty} \underbrace{d\xi \sqrt{\frac{\hbar}{m\omega}}}_{dx} \sqrt{\frac{\hbar}{m\omega}}\,\xi \left(e^{-\xi^2/2}\right)^2 =$$

$$= \frac{\hbar}{m\omega} \int_{-\infty}^{\infty} d\xi\, \xi \left(\xi e^{-\xi^2/2}\right)^2 = 0 \quad \text{by oddness.}$$

$$\hat{p} = \frac{\hbar}{i}\frac{d}{dx} = \frac{\hbar}{i}\frac{d}{d\xi}\frac{d\xi}{dx} = \frac{\hbar}{i}\sqrt{\frac{m\omega}{\hbar}}\frac{d}{d\xi}$$

But $\langle p\rangle_0 = \int d\xi \sqrt{\frac{\hbar}{m\omega}}\, \psi_0 \left(\frac{\hbar}{i}\sqrt{\frac{m\omega}{\hbar}}\frac{d}{d\xi}\right)\psi_0 =$

$$= \frac{\hbar}{i}\int d\xi\, \psi_1 \left(\frac{d}{d\xi}\right)\psi_1 = 0$$ because of the oddness of the integrand.

$$\langle x^2\rangle_n = \frac{\hbar}{m\omega}\int dx\, \xi^2 |\psi_n|^2 = \left(\frac{\hbar}{m\omega}\right)^{3/2}\int d\xi\, \xi^2 |\psi_n(\xi)|^2$$

For $\underline{n=0}$: $\langle x^2\rangle_0 = \alpha^2\left(\frac{\hbar}{m\omega}\right)^{3/2} \underbrace{\int_{-\infty}^{\infty} d\xi\, \xi^2 e^{-\xi^2}}_{\frac{\sqrt{\pi}}{2}} =$ using $\tilde{J}(\beta=1)$ on page 4.

$$= \frac{\sqrt{\pi}}{2}\left(\frac{m\omega}{\pi\hbar}\right)^{1/2}\left(\frac{\hbar}{m\omega}\right)^{3/2} = \underline{\frac{\hbar}{2m\omega}}$$

For $\underline{n=1}$: $\langle x^2\rangle_1 = (\sqrt{2}\alpha)^2\left(\frac{\hbar}{m\omega}\right)^{3/2}\int_{-\infty}^{\infty} d\xi\, \xi^2 \left(\xi e^{-\xi^2/2}\right)^2 =$

$$= 2\alpha^2\left(\frac{\hbar}{m\omega}\right)^{3/2} \underbrace{\int_{-\infty}^{\infty} d\xi\, \xi^4 e^{-\xi^2}} = \frac{3\sqrt{\pi}}{2}\left(\frac{m\omega}{\pi\hbar}\right)^{1/2}\left(\frac{\hbar}{m\omega}\right)^{3/2} = \underline{\frac{3\hbar}{2m\omega}}$$

$$\frac{d^2}{d\beta^2}\int d\xi\, e^{-\beta\xi^2}\Big|_{\beta=1} = \frac{d}{d\beta}\left(-\frac{\sqrt{\pi}}{2\beta^{3/2}}\right) = -\frac{\sqrt{\pi}}{2}\left(-\frac{3}{2}\right)\beta^{-5/2}\Big|_{\beta=1} = \frac{3\sqrt{\pi}}{4}.$$

Then $\hat{p}^2 = -\hbar^2\frac{d^2}{dx^2} = -\hbar^2\left(\frac{m\omega}{\hbar}\right)\frac{d^2}{d\xi^2}$

$$\Rightarrow \quad \langle p^2\rangle_n = -\hbar m\omega \int_{-\infty}^{\infty} \psi_n^*(\xi)\, \frac{d^2}{d\xi^2}\psi_n(\xi)\, \overbrace{d\xi \cdot \sqrt{\frac{\hbar}{m\omega}}}^{dx}$$

Find $\frac{d^2}{d\xi^2} e^{-\xi^2/2} = \frac{d}{d\xi}\left(-\xi e^{-\xi^2/2}\right) = -e^{-\xi^2/2} + \xi^2 e^{-\xi^2/2} = (\xi^2-1)e^{-\xi^2/2}$

$$\frac{d^2}{d\xi^2}\xi e^{-\xi^2/2} = \frac{d}{d\xi}\left(e^{-\xi^2/2} - \xi^2 e^{-\xi^2/2}\right) = -3\xi e^{-\xi^2/2} + \xi^3 e^{-\xi^2/2} =$$

$$= (\xi^3 - 3\xi)e^{-\xi^2/2} = \xi(\xi^2-3)e^{-\xi^2/2}$$

Thus $\langle p^2\rangle_0 = -\hbar m\omega\alpha^2 \int_{-\infty}^{\infty} e^{-\xi^2/2}(\xi^2-1)e^{-\xi^2/2}\left(d\xi \sqrt{\frac{\hbar}{m\omega}}\right)^{=dx} =$

$$= -\hbar m\omega\alpha^2\sqrt{\frac{\hbar}{m\omega}}\int_{-\infty}^{+\infty}(\xi^2-1)e^{-\xi^2}d\xi = -\hbar m\omega\alpha^2\left(\frac{\sqrt{\pi}}{2} - \sqrt{\pi}\right)\sqrt{\frac{\hbar}{m\omega}} =$$

$$= \hbar m\omega\left(\frac{m\omega}{\pi\hbar}\right)^{1/2}\frac{\sqrt{\pi}}{2}\cdot\sqrt{\frac{\hbar}{m\omega}} = \frac{\hbar m\omega}{2} \quad (\text{for } n=0)$$

$$\langle p^2\rangle_1 = -\hbar m\omega\sqrt{\frac{\hbar}{m\omega}}\left(\sqrt{2}\alpha\right)^2\int \xi e^{-\xi^2/2}\,\xi(\xi^2-3)e^{-\xi^2/2}d\xi =$$

$$= -2\alpha^2\sqrt{\hbar^3 m\omega}\int_{-\infty}^{\infty}\xi^2(\xi^2-3)e^{-\xi^2}d\xi$$

From $\tilde{J}(\beta) = \frac{\sqrt{\pi}}{2\beta^{3/2}} = \int_{-\infty}^{\infty} d\xi\, \xi^2 e^{-\beta\xi^2}$ we can find

that $\int d\xi\, \xi^4 e^{-\beta\xi^2} = -\frac{d}{d\beta}\tilde{J}(\beta)\Big|_{\beta=1} = \frac{\sqrt{\pi}}{2}\frac{3}{2}\beta^{-5/2}\Big|_{\beta=1} = \frac{3\sqrt{\pi}}{4}$

Thus $\langle p^2\rangle_1 = -2\alpha^2\sqrt{\hbar^3 m\omega}\left(\frac{3\sqrt{\pi}}{4} - 3\frac{\sqrt{\pi}}{2}\right) = 2\alpha^2\sqrt{\hbar^3 m\omega}\,\frac{3\sqrt{\pi}}{4} =$

$$= \frac{3\sqrt{\pi}}{2}\left(\frac{m\omega}{\pi\hbar}\right)^{1/2}\left(\hbar^3 m\omega\right)^{1/2} = \frac{3\hbar m\omega}{2} \quad (\text{for } n=1)$$

(b) For $\underline{n=0}$: $\sigma_x = \sqrt{\frac{\hbar}{2m\omega}}$ $\quad \sigma_p = \sqrt{\frac{\hbar m\omega}{2}} \Rightarrow \sigma_x\sigma_p = \frac{\hbar}{2}$ (minimal possible value!)

For $\underline{n=1}$: $\sigma_x = \sqrt{\frac{3\hbar}{2m\omega}}$ $\quad \sigma_p = \sqrt{\frac{3\hbar m\omega}{2}} \Rightarrow \sigma_x\sigma_p = \frac{3\hbar}{2}$.

(c) $\langle T \rangle = \langle \frac{\hat{p}^2}{2m} \rangle = \frac{\langle p \rangle^2}{2m} = \begin{cases} \frac{\hbar\omega}{4} & \text{for } n=0 \\ \frac{3\hbar\omega}{4} & \text{for } n=1 \end{cases}$

$\langle V \rangle = \langle \frac{1}{2} m\omega^2 x^2 \rangle = \frac{1}{2} m\omega^2 \langle x^2 \rangle = \begin{cases} \frac{\hbar\omega}{4} & \text{for } n=0 \\ \frac{3\hbar\omega}{4} & \text{for } n=1 \end{cases}$

$E = \langle T \rangle + \langle V \rangle = \begin{cases} \frac{1}{2}\hbar\omega & \text{for } n=0 \\ \frac{3}{2}\hbar\omega & \text{for } n=1 \end{cases}$

as it should be, of course, because

$E_n = \hbar\omega\left(n+\frac{1}{2}\right)$ for <u>n-th</u> level.

P2.13 $\Psi(x,0) = A\left(3\psi_0(x) + 4\psi_1(x)\right)$

(a) $$1 = \int_{-\infty}^{\infty} |\Psi(x,0)|^2 dx = |A|^2 \int \left(3\psi_0 + 4\psi_1\right)^* \left(3\psi_0 + 4\psi_1\right) dx$$

$$= |A|^2 \left\{ 9 \int |\psi_0|^2 dx + 16 \int |\psi_1|^2 dx + 12 \underbrace{\int \psi_0^* \psi_1 dx}_{=0} + 12 \underbrace{\int \psi_1^* \psi_0 dx}_{=0} \right\}$$

by orthogonality of $\{\psi_n\}$ set.

$$= 25|A|^2 \Rightarrow A = \frac{1}{5}$$

$$\Psi(x,0) = \frac{3}{5}\psi_0 + \frac{4}{5}\psi_1 . \qquad E_0 = \frac{1}{2}\hbar\omega, \; E_1 = \frac{3}{2}\hbar\omega$$

(b) $$\Psi(x,t) = \frac{3}{5}\psi_0(x)\, e^{-iE_0 t/\hbar} + \frac{4}{5}\psi_1(x)\, e^{-iE_1 t/\hbar} =$$

$$= \frac{3}{5}\psi_0\, e^{-i\omega t/2} + \frac{4}{5}\psi_1\, e^{-i3\omega t/2}$$

$$|\Psi(x,t)|^2 = \left(\frac{3}{5}\psi_0^* e^{i\omega t/2} + \frac{4}{5}\psi_1^* e^{i3\omega t/2}\right)\left(\frac{3}{5}\psi_0\, e^{-i\omega t/2} + \frac{4}{5}\psi_1 e^{-i3\omega t/2}\right)$$

$$= \frac{1}{25}\left(9|\psi_0|^2 + 16|\psi_1|^2 + 12\psi_0^*\psi_1 e^{-i\omega t} + 12\psi_1^*\psi_0 e^{i\omega t}\right)$$

take into account reality of ψ_n.

$$= \frac{1}{25}\left(9\psi_0^2 + 16\psi_1^2 + 24\psi_0\psi_1 \cos\omega t\right).$$

(c) $$\langle x \rangle = \frac{1}{25}\int_{-\infty}^{\infty} x\left(9\psi_0^2 + 16\psi_1^2 + 24\psi_0\psi_1\cos\omega t\right) dx$$

But $\int x\,\psi_0^2\, dx = \int x\,\psi_1^2\, dx = 0$

$$\Rightarrow \langle x \rangle = \frac{24}{25}\cos\omega t \int \sqrt{\frac{\hbar}{m\omega}}\,\xi\, \alpha e^{-\xi^2/2}\, \sqrt{2}\,\alpha\,\xi\, e^{-\xi^2/2} \sqrt{\frac{\hbar}{m\omega}}\, d\xi \Rightarrow$$

$$x = \sqrt{\frac{\hbar}{m\omega}}\,\xi, \quad \alpha = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}$$

$$\langle x\rangle = \frac{24}{25}\frac{\hbar}{m\omega}\sqrt{2}\,\alpha^2 \underbrace{\int \xi^2 e^{-\xi^2} d\xi}_{\frac{\sqrt{\pi}}{2}} \cos\omega t =$$

$$= \frac{24}{25}\frac{\hbar}{m\omega}\sqrt{\frac{\pi}{2}}\left(\frac{m\omega}{\pi\hbar}\right)^{1/2}\cos\omega t = \frac{24}{25}\sqrt{\frac{\hbar}{2m\omega}}\cos\omega t$$

$$\langle p\rangle = m\frac{d\langle x\rangle}{dt} = -\frac{24}{25}\sqrt{\frac{\hbar m\omega}{2}}\sin\omega t$$

The Ehrenfest's Theorem (P1.7) says $\frac{d\langle p\rangle}{dt} = \left\langle -\frac{\partial V}{\partial x}\right\rangle$

But $\frac{d\langle p\rangle}{dt} = -\frac{24}{25}\sqrt{\frac{\hbar m\omega}{2}}\,\omega\cos\omega t$

$$\left\langle -\frac{\partial V}{\partial x}\right\rangle = -\left\langle \frac{d}{dx}\frac{1}{2}m\omega^2x^2\right\rangle = -m\omega^2\langle x\rangle =$$

$$= -m\omega^2\cdot\frac{24}{25}\sqrt{\frac{\hbar}{2m\omega}}\cos\omega t = -\frac{24}{25}\sqrt{\frac{\hbar m\omega}{2}}\cdot\omega\cdot\cos\omega t$$

Thus $\frac{d\langle p\rangle}{dt} = -\frac{24}{25}\sqrt{\frac{\hbar m\omega}{2}}\,\omega\cos\omega t = \left\langle -\frac{\partial V}{\partial x}\right\rangle$ indeed.

(d) From $\Psi(x,t) = \frac{3}{5}\psi_0 e^{-iE_0t/\hbar} + \frac{4}{5}\psi_1 e^{-iE_1t/\hbar}$

follows that one can get $E_0 = \frac{1}{2}\hbar\omega$ with probability $P_0 = \left(\frac{3}{5}\right)^2 = \frac{9}{25}$.

and $E_1 = \frac{3}{2}\hbar\omega$ with $P_1 = \left|\frac{4}{5}\right|^2 = \frac{16}{25}$.